

$$\frac{p_{(k)}}{p} \leq \frac{k}{m_0 + j - 1} \frac{m_0 + j - 1}{(m + 1)p} q^*. \quad (7)$$

When conditioning on $P'_{(m_0)} = p$, the P'_i/p for $i = 1, 2, \dots, m_0 - 1$ are distributed as $m_0 - 1$ independent $U(0, 1)$ random variables, and the p_i/p for $i = 1, 2, \dots, j$ are numbers corresponding to false null hypotheses between 0 and 1. Using inequality (7) to test the $m_0 + j - 1 = m' \leq m$ hypotheses is equivalent to using procedure (1), with the constant $\{(m_0 + j - 1)/(m + 1)p\}q^*$ taking the role of q^* . Applying now the induction hypothesis, we have

$$E(Q | P'_{(m_0)} = p, P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) \leq \frac{m_0 - 1}{m_0 + j - 1} \frac{m_0 + j - 1}{(m + 1)p} q^* = \frac{m_0 - 1}{(m + 1)p} q^* \quad (8).$$

The bound in inequality (8) depends on p , but not on the segment $p_j < p < p_{j+1}$ for which it was evaluated, so

$$\begin{aligned} \int_{p^*}^1 E(Q | P'_{(m_0)} = p, P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) f'_{P'_{(m_0)}}(p) dp &\leq \int_{p^*}^1 \frac{m_0 - 1}{(m + 1)p} q^* m_0 p^{(m_0-1)} dp \\ &= \frac{m_0}{m + 1} q^* \int_{p^*}^1 (m_0 - 1) p^{(m_0-2)} dp = \frac{m_0}{m + 1} q^* \{1 - p^{(m_0-1)}\}. \end{aligned} \quad (9)$$

Adding inequalities (6) and (9) completes the proof of the lemma.

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